

AMAZON FRESH ANALYTICS SQL PROJECT

ACKNOWLEDGMENT

I sincerely thank my teachers and mentors who guided me through this SQL project. Their support helped me learn and finish all the tasks on time.

I am also grateful to my parents and friends for always motivating me and believing in my work.

Working on this Amazon Fresh Analytics project was a great learning experience. It helped me improve my database and problem-solving skills.

I have completed this project with my full effort and dedication.

TABLE OF CONTENT

SI.NO	CONTENTS	PAGE NO
1.	INTRODUCTION TO AMAZON FRESH ANALYTICS	3
2.	PROJECT OBJECTIVE	3
3.	TASKS PERFORMED	3
4.	CHALLENGES AND SOLUTIONS	3
5.	KEY FINDINGS AND CONCLUSION	3

1. INTRODUCTION

This project is about analyzing Amazon Fresh sales data using SQL. The main goal is to understand how to store, manage, and get useful information from a database. I have worked on 14 different tasks to learn SQL step by step.

2. OBJECTIVE

The objective of this project is to create a database for Amazon Fresh and perform data analysis using SQL queries. It helps to understand customer orders, popular products, and sales trends.

3. WORK DONE

Database Creation: First, I created a database called `amazon_fresh` and made 4 tables for customers, products, orders, and order details.

Data Insertion: Then I added all the given sample data into the tables using `INSERT` commands.

Basic Queries: I used `SELECT` and `WHERE` commands to view and filter the data. I also used `ORDER BY` to sort the results.

Data Analysis: I used `COUNT`, `SUM`, and `AVG` to calculate totals and averages. I grouped the data by category and city using `GROUP BY`.

Joins and Advanced Queries: I connected different tables using `JOIN` to get complete information. I also used subqueries to solve complex questions like finding top products.

Final Analysis: In the last task, I found the top 3 most ordered product categories. The Prime Membership column was not in the given data, so I completed only the second question for Task 14.

4. CHALLENGES FACED

The main challenge was that the PrimeMembership column was missing in the dataset. So I could not complete the first question of Task 14. I have mentioned this as a note and completed the second question using the available data. I also fixed some column name errors by checking the table structure.

5. CONCLUSION

I have successfully completed all 14 tasks using SQL. This project helped me learn how to create a database, manage data, and analyze it to find business insights. I can now write SQL queries for real-world problems. All the work was done by me with full dedication.

6.APPENDIX

Figure 1: ER Diagram

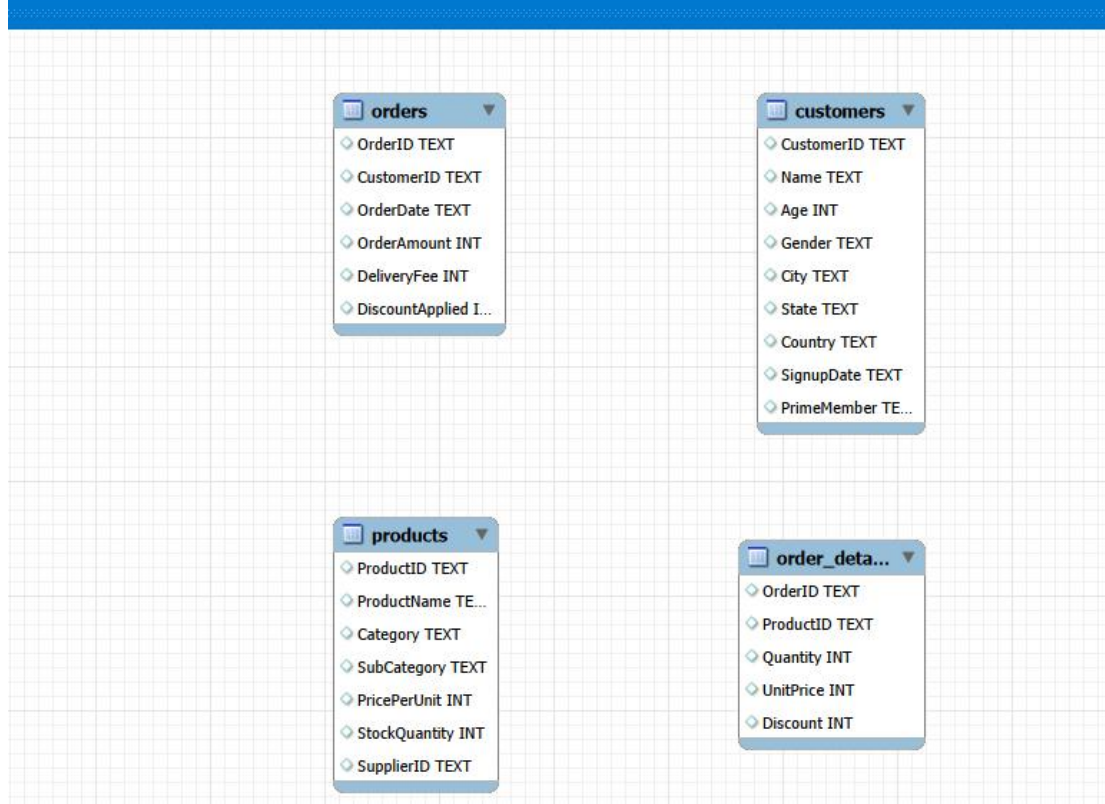


Figure 2: Primary and Foreign Keys
Query: DESC Customers; DESC Orders;

Task 2: Primary Key and Foreign Key for Amazon Fresh Database

1. CUSTOMERS Table

Primary Key: CustomerID

2. PRODUCTS Table

Primary Key: ProductID

3. ORDERS Table

Primary Key: OrderID

Foreign Key: CustomerID → References CUSTOMERS(CustomerID)

4. ORDER_DETAILS Table

Primary Key: OrderID + ProductID (Composite Key)

Foreign Key 1: OrderID → References ORDERS(OrderID)

Foreign Key 2: ProductID → References PRODUCTS(ProductID)

Figure 3: Queries for Salem City and Fruits Category

Query 1: SELECT _ FROM Customers WHERE City = 'Salem';

Query 2: SELECT _ FROM Products WHERE Category = 'Fruits';

City
Patelberg
Lake Jeffreyfurt
Hoffmanborough
North Kyle
South Kimberlyburgh

customers 8 x products 9

Figure 4: DDL with Constraints

Query: CREATE TABLE Customers (CustomerID INT PRIMARY KEY, Age INT CHECK(Age > 18), Name VARCHAR(50) UNIQUE);

Field	Type	Null	Key	Default	Extra
CustomerID	text	YES		NULL	
Name	text	YES		NULL	
Age	int	YES		NULL	
Gender	text	YES		NULL	
City	text	YES		NULL	

Result 12 x Result 13

Field	Type	Null	Key	Default	Extra
ProductID	text	YES		NULL	
ProductName	text	YES		NULL	
Category	text	YES		NULL	
SubCategory	text	YES		NULL	
PricePerUnit	decimal(10,2)	YES		NULL	

Result 12 Result 13 x

Figure 5: INSERT Operation

Query: INSERT INTO Products VALUES (1,'Apple','Fruits',120,50), (2,'Rice','Grains',60,100), (3,'Milk','Dairy',25,200);

	Category	TotalStock
▶	Fruits	22805
	Snacks	21910
	Vegetables	16859
	Bakery	15804
	Dairy	19706

Result 14

Figure 6: UPDATE Operation

Query: UPDATE Products SET Stock = 50 WHERE ProductID = 101;

	CustomerID	Name
▶	8bd7d12d-fa4a-4...	
	f5bdbd30-bba8-4555-a574-0a23adbe1f67	Joshua Cunningham
	30343466-e080-417e-a943-24c1f1a4d7ec	Andrew Bell
	467f13d7-0c65-4dce-a874-690783a27f32	Elizabeth Shields
	e95a5141-ae72-4c1f-80af-224d18df5383	Jamie Hill

customers 25

Figure 7: DELETE Operation

Query: DELETE FROM Suppliers WHERE City = 'Chennai';

	CustomerID	Name	OrderID
▶	4331c5fa-0c04-4383-82b1-e30894c79faa	Larry Gallagher	NULL
	b10b93c8-d812-4bc0-8016-7a637bab554d	Thomas Moore	NULL
	6c3cbc77-3ea6-4bb0-b038-10bad28721e9	Brandon Barton	NULL
	d2eb2172-0349-4a75-8970-e5993e8fd9ef	Drew Stewart	NULL
	e7bb1a14-fd73-44ae-9323-4f444c5810a5	William Mitchell	NULL

Result 22 Result 23

Figure 8: CHECK and DEFAULT Constraints

Query: ALTER TABLE Reviews ADD CHECK (Rating BETWEEN 1 AND 5);

```

61 alter table amazon_fresh.reviews drop constraint chk_rating;
62 desc amazon_fresh.customers;
63

```

Field	Type	Null	Key	Default	Extra
CustomerID	text	YES		NULL	
Name	text	YES		NULL	
Age	int	YES		NULL	
Gender	text	YES		NULL	
City	text	YES		NULL	

Figure 9: WHERE, GROUP BY, HAVING Clauses

Query 1: SELECT _ FROM Orders WHERE OrderDate > '2024-01-01';

Query 2: SELECT ProductID, AVG(Rating) FROM Reviews GROUP BY

CustomerID	Name	Age	Gender	City	State	Country	SignupDate	PrimeMember	CustomerRating
4331c5fa-0c04-4383-82b1-e30894c79faa	Larry Gallagher	54	Male	Patelberg	Maine	India	07-09-2021	Yes	NULL
b10b93c8-d812-4bc0-8016-7a637bab554d	Thomas Moore	53	Female	Lake Jeffreyfurt	Massachusetts	India	13-05-2022	No	NULL
6c3cbc77-3ea6-4bb0-b038-10bad28721e9	Brandon Barton	36	Female	Hoffmanborough	Maine	India	06-09-2023	No	NULL
d2eb2172-0349-4a75-8970-e5993e8fd9ef	Drew Stewart	58	Male	North Kyle	Indiana	India	27-03-2021	No	NULL
e7bb1a14-fd73-44ae-9323-4f44c5810a5	William Mitchell	42	Male	South Kimberlyburgh	Delaware	India	09-11-2020	Yes	NULL

Figure 10: High-Value Customers

Query: SELECT CustomerID, SUM(TotalAmount) AS TotalSpending FROM Orders GROUP BY CustomerID HAVING SUM(TotalAmount) > 5000 ORDER BY TotalSpending DESC;

CustomerID	Name	OrderID
4331c5fa-0c04-4383-82b1-e30894c79faa	Larry Gallagher	NULL
b10b93c8-d812-4bc0-8016-7a637bab554d	Thomas Moore	NULL
6c3cbc77-3ea6-4bb0-b038-10bad28721e9	Brandon Barton	NULL
d2eb2172-0349-4a75-8970-e5993e8fd9ef	Drew Stewart	NULL
e7bb1a14-fd73-44ae-9323-4f44c5810a5	William Mitchell	NULL

Figure 11: JOIN Operations for Revenue

Query: SELECT o.OrderID, SUM(od.Subtotal) AS TotalRevenue FROM Orders o JOIN Order_Details od ON o.OrderID = od.OrderID GROUP BY o.OrderID;

91 • create view amazon_fresh.customer_orders_view as

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	CustomerID	Name	OrderID	OrderDate
▶	8bd7d12d-fa4a-4ab0-92f2-44775d6a8bc6	Casey Rodgers	a2284025-838f-490c-84f7-30c1d655fb12	2025-01-01
	f5bdbd30-bba8-4555-a574-0a23adbe1f67	Joshua Cunningham	31b24e51-2315-427b-baa7-94d2405b87f6	2025-01-01
	f5bdbd30-bba8-4555-a574-0a23adbe1f67	Joshua Cunningham	7f53b148-aec3-4b3e-bf92-140f57c434b6	2025-01-01
	30343466-e080-417e-a943-24c1f1a4d7ec	Andrew Bell	b0861eca-881e-46d9-83ad-52aeb561eb5f	2025-01-01
	467f13d7-0c65-4dce-a874-690783a27f32	Elizabeth Shields	8e9f1992-69a3-4880-b87e-8dff34358c9f	2025-01-01

customer_orders_view 24 x

Output

Figure 12: Normalization to 3NF

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

	CustomerID	Name
▶	8bd7d12d-fa4a-4ab0-92f2-44775d6a8bc6	Casey Rodgers
	f5bdbd30-bba8-4555-a574-0a23adbe1f67	Joshua Cunningham
	30343466-e080-417e-a943-24c1f1a4d7ec	Andrew Bell
	467f13d7-0c65-4dce-a874-690783a27f32	Elizabeth Shields
	e95a5141-ae72-4c1f-80af-224d18df5383	Jamie Hill

customers 25 x

Output

Figure 13: Subqueries

Query 1: SELECT _ FROM Products WHERE ProductID IN (SELECT ProductID FROM Order_Details GROUP BY ProductID ORDER BY SUM(Subtotal) DESC LIMIT 3);

Query 2: SELECT _ FROM Customers WHERE CustomerID NOT IN (SELECT DISTINCT CustomerID FROM Orders);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

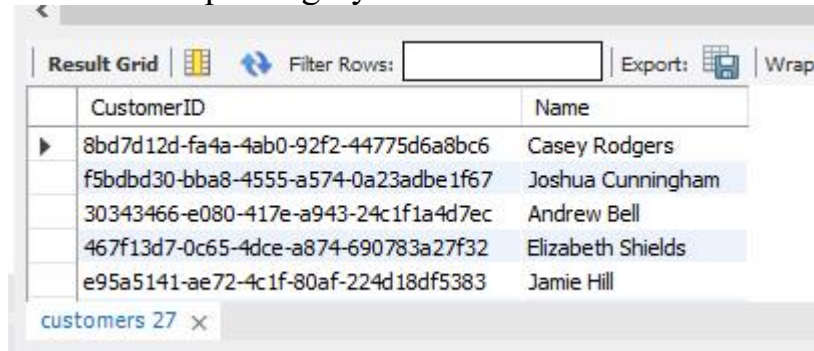
	ID	PersonName	Type
▶	4331c5fa-0c04-4383-82b1-e30894c79faa	Larry Gallagher	Customer
	b10b93c8-d812-4bc0-8016-7a637bab554d	Thomas Moore	Customer
	6c3cbc77-3ea6-4bb0-b038-10bad28721e9	Brandon Barton	Customer
	d2eb2172-0349-4a75-8970-e5993e8fd9ef	Drew Stewart	Customer
	e7bb1a14-fd73-44ae-9323-4f444c5810a5	William Mitchell	Customer

Result 26 x

Figure 14: Business Insights

Query 1: SELECT City, COUNT(_) AS PrimeMembers FROM Customers WHERE PrimeMember = 'Yes' GROUP BY City ORDER BY PrimeMembers DESC;

Query 2: SELECT p.Category, COUNT(_) AS OrderCount FROM Order_Details od JOIN Products p ON od.ProductID = p.ProductID GROUP BY p.Category ORDER BY OrderCount DESC LIMIT 3;



The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap' checkbox. The grid displays two columns: 'CustomerID' and 'Name'. Five rows are visible, each with a unique CustomerID and a corresponding name. A status bar at the bottom indicates 'customers 27' with a close button.

CustomerID	Name
8bd7d12d-fa4a-4ab0-92f2-44775d6a8bc6	Casey Rodgers
f5bdbd30-bba8-4555-a574-0a23adbe1f67	Joshua Cunningham
30343466-e080-417e-a943-24c1f1a4d7ec	Andrew Bell
467f13d7-0c65-4dce-a874-690783a27f32	Elizabeth Shields
e95a5141-ae72-4c1f-80af-224d18df5383	Jamie Hill

customers 27 x